Development of an Agentic AI Pipeline for Conducting Rigorous, Transparent, and Reproducible Systematic Reviews on Any Given Topic

1. Project Overview

This project aims to design, implement, and evaluate an autonomous multi-agent AI system that performs end-to-end systematic literature reviews. The system leverages LangChain for agent orchestration, tool use, and workflow management, combined with Retrieval-Augmented Generation (RAG) for grounded, evidence-based reasoning over large collections of scientific literature.

The system will accept a research question or topic as input and autonomously execute the full systematic review process following established methodological standards (e.g., PRISMA 2020 guidelines), including protocol definition, literature search, screening, data extraction, quality assessment, synthesis, and report generation.

2. Project Objectives

Build a modular, multi-agent architecture using LangChain that autonomously conducts systematic reviews.

Integrate advanced RAG pipelines for accurate retrieval and grounded generation from scientific corpora (PubMed, Semantic Scholar, arXiv, OpenAlex, etc.).

Implement full PRISMA-aligned workflow: protocol generation, search strategy design, dual screening simulation, data extraction, risk-of-bias assessment, qualitative/quantitative synthesis, and final report generation.

Ensure transparency through detailed logging, citation tracking, decision rationales, and reproducible execution traces.

Enable human-in-the-loop oversight at critical decision points while maximizing autonomy.

Deliver a general-purpose system that works for any given research topic with minimal configuration.

3. Scope

Topic/question intake and research protocol generation

Automated development of comprehensive search strategies (Boolean, MeSH, etc.)

Multi-source literature retrieval and deduplication

Title/abstract and full-text screening with inclusion/exclusion rationale

Structured data extraction and evidence tables

Quality / risk-of-bias assessment

Narrative synthesis and (where applicable) meta-analysis preparation

Generation of PRISMA flow diagram data, evidence maps, and final review manuscript draft

Persistent memory, versioning, and audit trails

4. Proposed Technical Approach

Architecture

Multi-agent system orchestrated via LangChain (or LangGraph for stateful workflows):

Coordinator / Planner Agent: Decomposes the review into stages and manages the overall workflow.

Search Strategy Agent: Designs and refines database queries.

Retrieval Agent: Executes searches and manages RAG index construction/updates.

Screening Agents (Title/Abstract + Full-text): Apply inclusion/exclusion criteria with reasoning.

Extraction & Appraisal Agent: Performs structured data extraction and quality assessment.

Synthesis & Writing Agent: Produces narrative synthesis, tables, and the final report.

Critic / Quality Assurance Agent: Reviews outputs for consistency, bias, completeness, and adherence to protocol.

Core Technologies

LangChain / LangGraph: Agent definition, tool calling, memory, routing, and state management.

RAG Pipeline: Hybrid retrieval (dense + sparse), re-ranking, and context compression over vector stores (e.g., Chroma, FAISS, or Pinecone) populated from scientific sources.

LLM Backbone: High-capability models with strong reasoning and long-context support.

Tools & Integrations: Academic search APIs, PDF parsing, citation managers, PRISMA checklist validators, and export utilities (Markdown, DOCX, BibTeX).

Observability: Full tracing of agent decisions, retrieved documents, and intermediate artifacts.

5. Expected Deliverables

Fully functional agentic AI system (source code + documentation in Github/Gitlab

Configurable review protocol template aligned with PRISMA.

End-to-end demonstration on at least one real research topic.

Technical report and user guide (readme file or pdf).

Evaluation results measuring completeness, accuracy, inter-agent consistency, and time savings versus manual review.

6. Success Criteria

The system successfully completes a full systematic review pipeline with minimal human intervention.

Outputs are methodologically sound, fully referenced, and transparent.

Screening and extraction decisions include clear, traceable rationales.

Significant reduction in time and effort compared with traditional systematic review processes while maintaining or improving rigor.

Project Submission Requirements

1. Source Code: All project source code must be uploaded to GitHub or GitLab.
2. README Documentation: A README.md file must be provided, documenting all the steps followed during project development and setup, including installation, configuration, execution, and usage instructions.
3. Video Demonstration: A video demonstrating the implemented project must be submitted. The video must not exceed 3 minutes.
4. Teamwork and Individual Contributions: Students must work in teams of no more than five members. Each student must clearly describe their individual contribution to the project in the final report.